The Reason Why Some Distillation Models Could Not Use DDIMInverse

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1 Proof

In the following essay, I will first provide you with the most concise deduction of the ground-truth training target for the diffusion model, which doesn't require complex physics knowledge, such as the FP formula. And then, show you the reason why we could not apply DDIM as well as DDIM Inverse on the distillation model.

This is the most concise deduction I have ever seen. This deduction does not really require us to understand those complex skills for high-dimensional mathematical objects. The function that takes a vector as input can be deemed a multivariable function. The ∇ operator is used only to express the basic Taylor formula in a concise way.

So, let us start.

For arbitrary interpolation between image X_1 and noise X_0, we have the following non-causal trajectory in the training:

$$X_t = \varphi_t(X_1, X_0) = \alpha_t X_1 + \beta_t X_0 \tag{1}$$

Then, for those X_t in eq. (1), we have a conditional velocity use image X_1 as condition:

$$u^{cond}(X_t, X_1, t) = \frac{\partial \varphi_t}{\partial t} = \frac{-\dot{\beta}_t \alpha_t X_1 + \dot{\beta}_t X_t + \dot{\alpha}_t \beta_t X_1}{\beta_t} \tag{2}$$

, in which $\dot{\alpha}_t$ and $\dot{\beta}_t$ is the time derivatives of the α_t and β_t with respecti to t.

Although the conditional velocity described by eq. (2) provides a path of the eq. (1) to transport distribution, it is non-causal and can only be used in the training, since it requires the target x_1 in the future timestep, and, based on the Test Function Method, we could only use a neural network approximating the eq. (3) to do the inference, which is also the definition of the ground truth velocity field.

$$u^*(x_t, t) = \mathbb{E}_{x_1|x_t}[u^{cond}(x_t, x_1, t)] \tag{3}$$

Explicitly, for inference, if we have:

$$dZ_t = \mathbb{E}_{x_1|x_t}[u^{cond}(X_t, X_1, t)]dt \tag{4}$$

Then, we can transform $p_t(X_t)$ to $q_{t+\Delta t}(Z_{t+\Delta t})$ via applying the following formula eq. (5) to all possible X_t:

$$Z_{t+\Delta t} = X_t + \mathbb{E}_{x_1|x_t}[u^{cond}(X_t, X_1, t)]\Delta t \tag{5}$$

We use Z_t there to explicitly show that Z_t is given via neural network inference, while X_t is given by adding noise to a real image. In the real inference, we transport Z_t via eq. (4), but the following proof directly tells you that, if we apply eq. (5) to noisy X_t, we can get $Z_{t+\Delta t}$ that belongs to the distribution of adding noise to X_1 to get $X_{t+\Delta t}$. And starting from $Z_0 \sim \mathcal{N}(0, I)$, we can get a series of proper distributions.

Explicitly, based on the Test Function Method in eq. (6)

Theorem 1. *If for any test function $\phi(x)$, the following equation correct, then $f(x) = g(x)$:*

$$\int f(x)\phi(x)dx = \int g(x)\phi(x)dx \tag{6}$$

So, we need to prove that:

$$\int p_{t+\Delta t}(X_{t+\Delta t})\phi(X_{t+\Delta t})dX_{t+\Delta t} = \int q_{t+\Delta t}(Z_{t+\Delta t})\phi(Z_{t+\Delta t})dZ_{t+\Delta t} \tag{7}$$

,where X_t is given by non-causal interpolation, while Z_t given by the eq. (5).

Since for each given X_t, the ground truth velocity field $u^*(X_t, t)$ is deterministic. For the right part of the eq. (7), we can prevent inergrating $Z_{t+\Delta t} \sim q_{t+\Delta t}(Z_{t+\Delta t})$ but directly integrate $X_t \sim p_t(X_t)$.

and our proving target becomes:

$$\int p_{t+\Delta t}(X_{t+\Delta t})\phi(X_{t+\Delta t})dX_{t+\Delta t} = \int p_t(X_t)\phi(Z_{t+\Delta t})dX_t \tag{8}$$

Then, we simply rewrite it as eq. (9):

$$\int p_{t+\Delta t}(x_{t+\Delta t})\phi(x_{t+\Delta t})dx_{t+\Delta t} = \int p_t(x_t)\phi(x_t + u^*(x_t, t)\Delta t)dx_t \tag{9}$$

This can be proven by the following deduction:

$$\begin{aligned}
\mathbb{E}_{X_{t+\Delta t}}[\phi(X_{t+\Delta t})] &= \mathbb{E}_{X_0,X_t}[\phi(\varphi_{t+\Delta t})] \\
&= \mathbb{E}_{X_0,X_1}[\phi(\varphi_t) + \Delta t\frac{\partial \varphi_t}{\partial t} \cdot \nabla_{\varphi_t}\phi(\varphi_t)] \\
&= \mathbb{E}_{X_t}[\phi(X_t)] + \Delta t\mathbb{E}_{X_0,X_1}[\frac{\partial \varphi_t}{\partial t}\nabla_{X_t}\phi(X_t)]
\end{aligned} \tag{10}$$

where:

$$\nabla_{\varphi_t} = [\frac{\partial}{\partial X_t^0}, \frac{\partial}{\partial X_t^1}\frac{\partial}{\partial X_t^0}, ..., \frac{\partial}{\partial X_t^d}]^T \tag{11}$$

and,

$$\nabla_{\varphi_t}\phi(\varphi_t) = [\frac{\partial \phi(\varphi_t)}{\partial X_t^0}, \frac{\partial \phi(\varphi_t)}{\partial X_t^1}\frac{\partial \phi(\varphi_t)}{\partial X_t^0}, ..., \frac{\partial \phi(\varphi_t)}{\partial X_t^d}]^T \tag{12}$$

More explicitly, $\varphi_{t+\Delta t}(X_1, X_0)$ given by the eq. (1) which outputs a vector $X_t \in \mathbb{R}^d$, and the '$\cdot$' in the second equation of the eq. (10) is the inner product of two vector. The second equation is given by the first-order Taylor's formula. Meanwhile, the third equation substitute φ_t by X_t because once given X_1, X_0, the X_t is deterministic.

if x_1 in the eq. (2) can be deduce by given x_0, x_t, then we have:

$$\begin{aligned}
\mathbb{E}_{X_{t+\Delta t}}[\phi(X_{t+\Delta t})] &= \mathbb{E}_{X_t}[\phi(X_t)] + \Delta t \mathbb{E}_{X_0, X_t}[\frac{\partial \varphi_t}{\partial t}\nabla_{X_t}\phi(X_t)] \\
&= \mathbb{E}_{X_t}[\phi(X_t)] + \Delta t \mathbb{E}_{X_t}[\mathbb{E}_{X_1|X_t}[u^{cond}(x_t, t)]\nabla_{X_t}\phi(X_t)] \\
&= \mathbb{E}_{X_t}[\phi(X_t + \Delta t \mathbb{E}_{X_1|X_t}[u^{cond}(x_t, t)])] \\
&= \mathbb{E}_{X_t}[\phi(X_t + \Delta t u^*(x_t, t))]
\end{aligned} \tag{13}$$

So, we prove, for all of the function $\phi(x)$, for arbitary distribution p_0, p_1 and for any interpolation using different α_t, β_t that $\alpha_1 = 1, \beta_1 = 0$, without considering the relationship $p(X_1|X_0)$, the eq. (9) is always correct, which means for X_t describe by the non-causal path φ_t, we could gain its corresponding distribution by letting $X_{t-\Delta t}$ add a ground truth velocity field $u^*(X_t, t)$. So, starting from pure Gaussian noise X_0, we can get the correct probability distribution in each step.

This proof is basically based on the deduction provided by Jianlin Su, the author of the ROPE. You can check it there if Google can help you translate it: https://spaces.ac.cn/archives/9497

Then, once the $p(X_1|X_0)$ changes, the velocity field given by eq. (3) also changes, and the transportation trajectory also changes. The rectified flow proves that, if we use a diffusion model to get a noisy-image pair and interpolate between that noise and that image to get a noisy image, then inputting the noisy image to the new NN and letting it study the new velocity given by the noise-image pair could provide us a new trajectory that is straighter.

During distillation model training, we tend to avoid multi-step inference to obtain the noisy image pair. But if we can get those noise-image pairs via a few-step inference, then we already have our target NN. That seems like a paradox.

To cope with this paradox, the Consistency Model uses the following training method: use the parameters of the pretrained NN to initialize the student model, use the EMA parameters to initialize the teacher model, then input less noisy latents to the teacher model, and let the student model output as consistently as possible to the teacher model. During training, keep using the EMA algorithm to update the teacher model.

Then, DMD uses the pretrained diffusion model (DM) to provide a special loss to supervise the training of the student model. Explicitly, it lets the DM input a noisy image whose original clean image is generated by the student model, and study the noise added to the image. And then compare the trained DM and pretrained DM's output to train the distillation model; if the trained DM shifts its output, that means the DM tries to study a new probability distribution, which is different from the probability distribution of the training image set (since the DM studies the score function $\nabla_{X_t} \log P(X_t)$ to model the probability$P(X_1)$). By using this method, the NN will try to directly output an image that constitutes a probability distribution that is identical to the original probability distribution.

Both the CM model and the DMD model can directly output an image when given Gaussian noise as input.

But if we need multi-step inference, another paradox appears now. To prevent multi-step inference before training, we normally would directly add noise to the image in the training set and input it to the neural network, but in such a situation, the next-step input in inference, if we use an ODE solver and a model that directly outputs an image, is not similar to the noisy image that we directly add noise to the training set.

More explicitly, in the previous deduction based on the test function method, the deduction doesn't require any additional assumption on $p(X_1|X_0)$. So, once $p(X_1|X_0)$ has been changed, the ground truth velocity field described by eq. (3) has also been changed.

And if $X_{t+\Delta t}$ is obtained by adding arbitrary noise to the image, then the test function method will show that $X_{t+\Delta t}$, obtained by an interpolation between X_1 and X_0 in which $p(X_1|X_0) = p(X_1)$, is not, by definition, indistinguishable from the $Z_{t+\Delta t}$ for which the noise image-pair has the relationship $p(X_1|X_0) = 1$.

So, if a researcher trains an RF-2 model, then it cannot input a probability distribution given by randomly adding noise to the real image. And, if a distillation model in the training randomly adds noise to the image and gets the input, it cannot use the output from the RF-2 model as input.

I really don't want to repeat the proof of the RF, which claims that after applying the RF algorithm, the trajectory is straighter. For DMD, I provide a basic idea there: If the ground truth trajectory is totally straight, the inference trajectory, in the RF algorithm, is equal to the training trajectory, and then 1-step inference is enough to generate an image, since the inference process is not a curve, and that situation only appears when it achieves the optimal transportation.

So, DMDs that tend to let the NN output a real image directly actually want to implement the RF-Inf directly.

So, if you use DMD to train a NN, then its first step output given by an ODE solver, a noisy image Z_{t_1}, is also different from the X_{t_1} that adds random noise directly to a real image. That conclusion, at least I believe, is still correct for all subsequent timesteps t. Hence, those distillation models should use the LCM solver, which adds noise to the clean image output by the distillation model.

In DMD, we are forced to use the LCM solver: it first outputs an image and

then perturbs it with random noise. In this setting, the resulting SDE may look superficially similar to the one implied by the Fokker–Planck (FP) equation, but it does not actually correspond to (i.e., solve) that FP-induced SDE. What, then, is the FP formula telling us? If we don't use the test-function proof, we can draw an additional conclusion: there exists a family of SDEs that generates noisy features whose distribution matches the distribution of the noisy features produced by ODE-based inference.

To prove it, I could provide you with a Python script. It shows the result of applying different solvers to the DMD model. If the LCM solver is a special SDE solver given by the FP formula, then DDIM should still work, but in practice, the perceptual quality obviously decreases.

Since the DMD model and LCM model cannot use the DDIM solver, they also cannot use the DDIM inverse solver.